Extrapancreatic Advanced Endoscopic Interventions

Haresh Vijay Naringrekar, MD
Haroon Shahid, MD
Cyril Varghese, MD
Alex Schlachterman, MD
Sandeep P. Deshmukh, MD
Christopher G. Roth, MD, MS

Abbreviations: EDGE = endoscopic US-directed transgastric endoscopic retrograde cholangiopancreatography, ESD = endoscopic submucosal dissection, ESG = endoscopic sleeve gastroplasty, EUS = endoscopic US, LAMS = lumen-apposing metal stent, POEM = peroral endoscopic myotomy

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From the Department of Radiology (H.V.N., S.P.D., C.G.R.) and Department of Gastroenterology and Hepatology (A.S.), Thomas Jefferson University Hospital, 132 S 10th St, Philadelphia, PA 19123; Division of Gastroenterology and Hepatology, Rutgers-Robert Wood Johnson Medical School, New Brunswick, NJ (H.S.); and Department of Radiology, Westchester Medical Center, Valhalla, NY (C.V.). Presented as an education exhibit at the 2020 RSNA Annual Meeting. Received April 1, 2021; revision requested May 28 and received June 16; accepted June 24. For this journal-based SA-CME activity, the author A.S. has provided disclosures (see end of article); all other authors, the editor, and the reviewers have disclosed no relevant relationships. **Address correspondence to** H.V.N. (e-mail: *haringrekar@gmail.com, hvn001@jefferson.edu*).

SA-CME LEARNING OBJECTIVES

After completing this journal-based SA-CME activity, participants will be able to:

- Identify the normal postoperative imaging appearance in patients who have undergone extrapancreatic endoscopic procedures.
- Describe extrapancreatic endoscopic procedures by using appropriate terminology.
- Recognize the potential complications from extrapancreatic endoscopic procedures.

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As the field of interventional endoscopy advances, conditions that were once treated with surgery are increasingly being treated with advanced endoscopy. Endoscopy is now used for treatment of achalasia, bariatric procedures for obesity; resection of early-stage malignancies in the gastrointestinal tract; and placement of lumen-apposing metal stents in the treatment of biliary obstruction, gastric outlet obstruction, cholecystitis, and drainage of nonpancreatic-related fluid collections or abscesses. Knowledge of the novel terminology, procedural details, expected postintervention imaging findings, and potential complications is vital for radiologists because these procedures are rapidly becoming more mainstream in daily practice. These procedures include peroral endoscopic myotomy for the treatment of achalasia and other esophageal motility disorders; endoscopic sleeve gastroplasty and placement of an intragastric balloon for weight loss; endoscopic submucosal dissection in the resection of tumors of the gastrointestinal tract; and therapeutic endoscopic–guided procedures for the treatment of biliary obstruction, gastric outlet obstruction, acute cholecystitis, and drainage of nonpancreatically related fluid collections. Patients benefit from these minimally invasive procedures, with potential improvement in morbidity and mortality rates, decreased length of hospital stay, and decreased health care costs when compared with the surgical alternative. Complications of these procedures include leaks or perforations, infections or abscesses, fistulas, and occlusion and migration of stents.

An invited commentary by Pisipati and Pannala is available online.

Introduction

The use of endoscopic techniques has been expanding rapidly, with extension into treatments of disease processes that have been traditionally performed surgically. These include treatment of pancreatitis and its complications, bariatric weight loss procedures, and endoluminal gallbladder drainage in patients with acute cholecystitis. Advantages of an endoluminal approach include a shorter hospital

TEACHING POINTS

- Many patients develop an intraluminal dissection, in which the mucosotomy is incompletely closed and demonstrates the passage of contrast material through the defect and into the submucosal space. This is a common finding, occurring in 30% of patients after the procedure, and it is most often clinically insignificant.
- Invasion into the muscularis propria or more advanced tumors seen at endoscopy or EUS may warrant further imaging for staging, including CT of the chest, abdomen, and pelvis and rectal MRI, depending on the location of the tumor.
- The gastric lumen appears constricted and tubular, sparing the fundus, which is flared outward and should not be misinterpreted as dehiscence or a suboptimal postprocedural appearance. Preserving the normal fundus also helps to prevent gastroesophageal reflux and enhances satiety by serving as a food reservoir.
- On all postprocedural images, the stent must be carefully assessed for subtle changes in placement that could suggest migration.
- A specific complication that is associated with the EDGE procedure after removal of the LAMS is a persistent patency of the fistula tract between the excluded stomach and gastric pouch, resulting in a gastrogastric fistula and recurrent weight gain.

stay, lower morbidity and mortality rates, and decreased health care costs (1). In addition, because of the minimally invasive nature of these techniques, patients in whom a surgical procedure would otherwise be contraindicated may be able to undergo the intervention endoscopically (2). As these procedures become more common in practice, radiologists must be able to identify the type of intervention performed, the expected postoperative imaging appearance, and the potential complications. This article is focused on several of these endoscopic interventions, including peroral endoscopic myotomy (POEM) in the treatment of achalasia, endoscopic sleeve gastroplasty (ESG) and placement of an intragastric balloon for treatment of obesity, and endoscopic submucosal dissection (ESD) for treatment of early-stage gastrointestinal tumors. In addition, we review the use of therapeutic endoscopic US (EUS) and placement of a lumen-apposing metal stent (LAMS) in patients with disease processes that are not related to the pancreas. The purposes of this article are to review the indications and advantages of these procedures, explain the endoscopic technique, familiarize radiologists with the expected imaging appearance of the postoperative anatomy, and describe the imaging findings of potential complications. Radiologic assessment of therapeutic endoscopic procedures for pancreatic disease is beyond the scope of this review and has been the subject of several review articles in the literature (3,4).

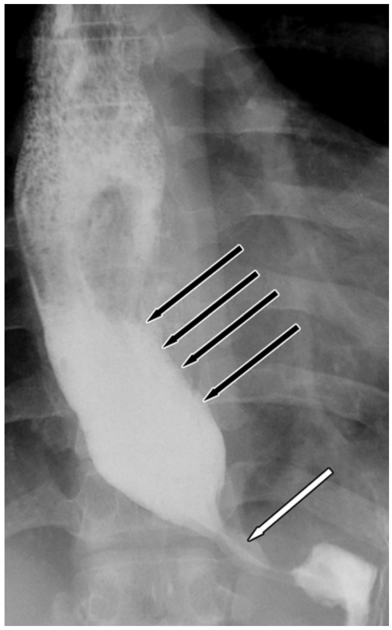


Figure 1. Achalasia in a 45-year-old man. Fluoroscopic image of the esophagus shows marked narrowing of the gastroesophageal junction (white arrow), with a dilated debris-filled esophagus (black arrows), which are findings consistent with achalasia.

Peroral Endoscopic Myotomy

Indications and Preliminary Workup

Achalasia is an esophageal motility disorder of unknown causes that affects the lower esophageal sphincter. It is a neuromuscular disorder that causes a loss of peristalsis and an inability of the gastroesophageal junction to relax, often resulting in a dilated esophagus. Achalasia is divided into three subtypes according to the results of high-resolution manometry. Symptoms include dysphagia, reflux, regurgitation, and weight loss (5). Overall, it is a rare motility disorder, occurring in approximately one individual per 100 000 persons each year (5,6). Preoperative workup for achalasia includes a barium swallow study, high-resolution esophageal manometry, and upper endoscopy (Fig 1). The classic imaging appearance of achalasia after barium swallow includes marked dilatation of the esophagus, with a bird-beaked narrowing at the gastroesophageal junction. Esophageal peristalsis is often markedly abnormal, with nonpropulsive tertiary contractions (7). Chest CT can be performed to assess for upper abdominal metastases such as infiltrating gastric cancer involving the gastroesophageal junction, which can manifest with similar symptoms. Known as pseudoachalasia, it is a result of tumor infiltration into the myenteric plexus and is the diagnosis for 2.4%–5.4% of patients with achalasia syndrome. Clinical findings suggestive of pseudoachalasia include

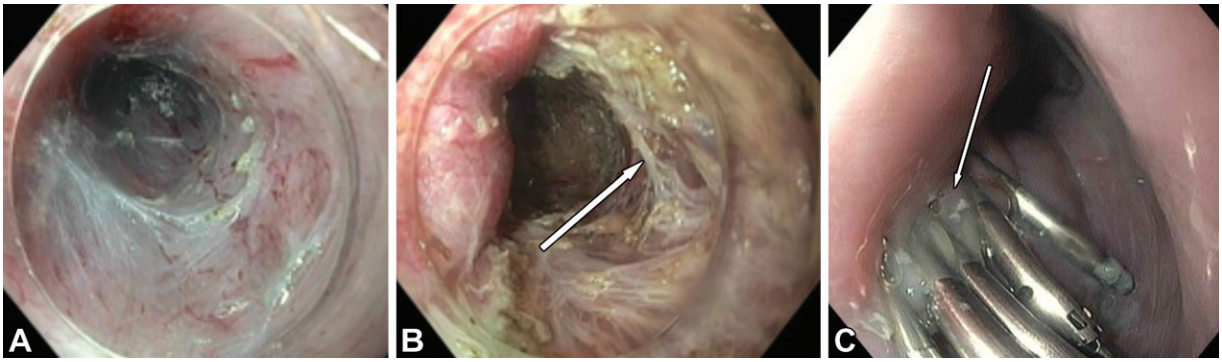


Figure 2. Achalasia in a 36-year-old man who underwent peroral endoscopic myotomy (POEM). **(A)** Endoscopic image shows a submucosal tunnel that was created endoscopically to gain access to the circumferential muscle fibers. **(B)** Endoscopic image shows that the muscle fibers (arrow) are cut to create the myotomy and release pressure at the lower esophageal sphincter. **(C)** Endoscopic image shows that the submucosal tunnel is closed with clips (arrow) and hemostasis is achieved. These clips remain after the procedure and pass spontaneously into the bowel several weeks later.

advanced patient age at presentation, acute onset of dysphagia, and substantial weight loss (8).

Endoscopic Technique

Treatment of achalasia is focused on relaxation of the esophagogastric junction and includes short-term treatments such as injection of botulinum toxin in the lower esophageal sphincter and long-term treatments such as pneumatic esophageal dilatation and laparoscopic Heller myotomy. Surgery has been the mainstay in treatment of achalasia, and laparoscopic Heller myotomy has been shown to be the most efficacious procedure, with reported success rates of 89% (9). This is achieved by disrupting the circumferential and longitudinal muscle fibers of the distal esophagus. POEM is a newer minimally invasive technique that is used in the treatment of achalasia and was developed in the last decade. The esophagus is insufflated with carbon dioxide during endoscopy, and mucosotomy is performed in the midesophagus to enter the submucosal space. The endoscope is advanced into the submucosal space, and a submucosal tunnel is then created between the muscularis propria and the mucosal layers. This is extended across the gastroesophageal junction and into the proximal stomach. Selective myotomy with division of the muscle fibers is performed by means of electrosurgery from the distal esophagus down to 2–3 cm distal to the gastroesophageal junction. The length of the myotomy is determined on the basis of the findings of high-resolution manometry (Fig 2) (5,10). Hemostasis is achieved, and the mucosotomy site is closed endoscopically with clips or sutures. The patient undergoes a follow-up barium swallow examination with water-soluble contrast material, followed by barium (if no extraluminal contrast material is initially seen) to ensure that there are no leaks and there is adequate esophageal emptying.

Expected Postoperative Appearance

A postprocedural swallow study demonstrates endoscopic clips at the midesophagus in the area of the submucosal tunnel. These clips remain after the procedure and often pass spontaneously after 2–3 weeks (7). Contrast material should pass readily through the gastroesophageal junction after the procedure, with mild residual luminal narrowing; a lumen diameter of 5 mm or greater at the distal esophagus has preliminarily been shown to be predictive of a good chance of success in patients after undergoing the POEM procedure (Fig 3) (11). Common postprocedural findings include pneumoperitoneum, pneumomediastinum, and retroperitoneal free air and are thought to be related to passage of carbon dioxide through small mucosal tears in the esophagus. Many patients develop an intraluminal dissection, in which the mucosotomy is incompletely closed and demonstrates the passage of contrast material through the defect and into the submucosal space. This is a common finding, occurring in 30% of patients after the procedure, and it is most often clinically insignificant (11). A barium swallow study shows this finding as a linear collection of contrast material, originating near the endoscopic clips and resolving within 3 weeks after the procedure (Fig 4).

Advantages and Complications

POEM has shown similar efficacy to that of laparoscopic Heller myotomy in the treatment of achalasia, with reported clinical success rates of 89%–100% (12). Advantages of POEM over laparoscopic Heller myotomy include decreased length of the procedure, decreased length of hospital stay, and decreased number of adverse events (10). Complications in patients after POEM are rare and include bleeding, esophageal

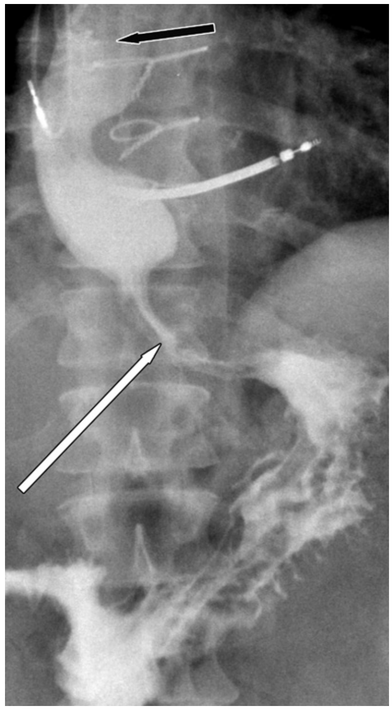


Figure 3. Normal appearance of the esophagus after POEM in a 42-year-old woman with achalasia. Fluoroscopic image shows the normal passage of contrast material through the gastroesophageal junction and into the stomach (white arrow). The diameter of the gastroesophageal junction measures larger than 5 mm, with no delay in passage of the contrast material, which is consistent with the expected postoperative appearance. Note the clips placed for hemostasis (black arrow), which pass spontaneously.

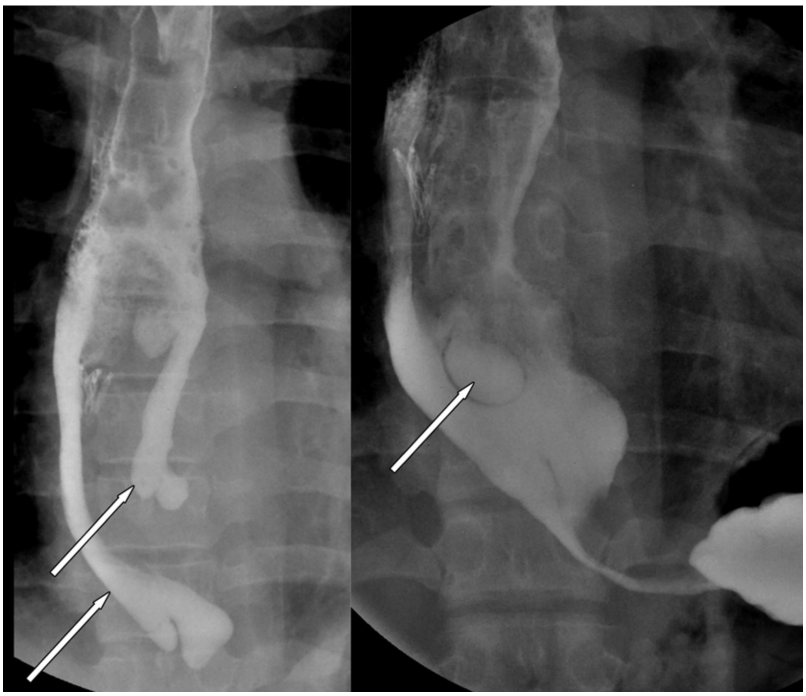


Figure 4. Intraluminal dissections in a 53-year-old man who was treated with POEM for achalasia. Anteroposterior fluoroscopic barium swallow study images show multiple intraluminal dissections (arrows). The patient was asymptomatic, and contrast material readily passed into the stomach with minimal delay. A subsequent barium swallow study performed the next day (not shown) showed a decreased size and number of intraluminal dissections.

leaks, gastric perforation, gastric inflammation, pleural effusion, and persistent or recurrent achalasia. Postoperative leaks appear as outpouchings beyond the expected lumen of the esophagus and are often irregularly marginated; these most commonly occur at the site of mucosal entry and should be distinguished from intraluminal dissections, which are a common postoperative finding and are most often inconsequential. Small leaks often resolve on their own, while larger leaks may require endoscopic repair or, potentially, surgical intervention. A large-volume pneumoperitoneum and gastric emphysema suggest perforation, which occurs more commonly at the gastric cardia (7). Occasionally, intraluminal dissections can become large enough to lead to obstruction at the gastroesophageal junction; this is extremely rare and is corrected by extending the myotomy further distally (Fig 5). Persistent achalasia after unsuccessful POEM is another potential complication and has been reported in up to 18% of patients (13).

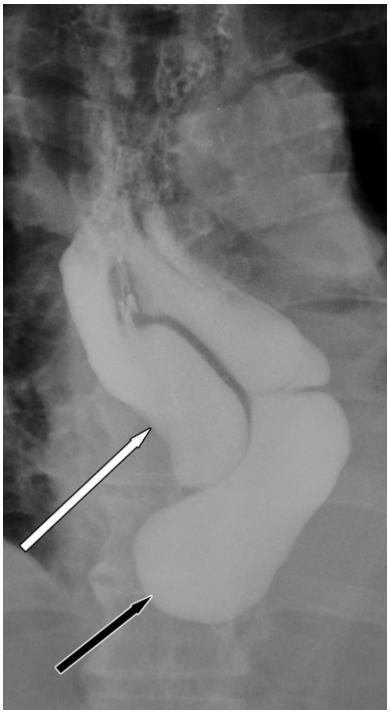


Figure 5. Large intraluminal dissection causing obstruction in a 31-year-old woman who underwent POEM. Barium swallow study image acquired after POEM shows a large intraluminal dissection (white arrow) and obstruction to the passage of contrast material at the gastroesophageal junction, with no contrast material passing into the stomach (black arrow). The patient was symptomatic and underwent extension of the myotomy. Barium swallow image acquired after extension of the myotomy (not shown) showed resolution of the intraluminal dissection, with contrast material passing into the stomach.

Endoscopic Submucosal Dissection

Indications and Preoperative Imaging

Traditionally, early gastrointestinal tumors in the esophagus, stomach, and colon have been treated surgically. Early tumors are defined by their confinement to the mucosa or submucosa, regardless of their size, or by local-regional involvement of the lymph nodes (2). ESD is a minimally invasive technique and has been used as an alternative to surgery in the resection of early gastrointestinal tumors. This endoscopic technique results in en-bloc resection of superficial tumors, allowing exact histologic evaluation and a low risk of recurrence (14). More specific indications for ESD are based on the location of the tumor. Esophageal ESD is typically reserved for patients with T1a disease, suspected superficial submucosal invasion of up to 100 μm, recurrent dysplasia, carcinoma with positive margins, and intramucosal carcinoma. Deep submucosal invasion (T1b-SM3) or the presence of regional adenopathy is a contraindication to ESD. Gastric ESD can be performed in patients with superficial mucosal tumors measuring 2 cm or smaller that are well differentiated and nonulcerated. The expanded criteria encompass tumors that measure 3 cm and are moderately to well differentiated. Colorectal ESD can be performed in lesions that are larger than 2 cm, with limited submucosal invasion and a nongranular shape and size or those that are larger than 3 cm, with granular or mixed morphologic characteristics (15). More expanded indications are given in Table 1 (16,17).

Preoperative imaging begins with high-resolution endoscopy, with use of narrow-band imaging or chromoendoscopy with Lugol solution as a contrast agent; these techniques allow the endoscopist to delineate the borders, size, and shape of the tumor (18). Tumors are accurately characterized with various classification systems including the Paris classification, pit pattern classification with the use of optical magnification, and narrow-band imaging classification systems (16). EUS can be used to assess the depth of tumor invasion or lymph node involvement in lesions at higher risk of malignancy. The accuracy of EUS for this purpose is variable, with 40%–60% of superficial lesions staged correctly (19). Invasion into the muscularis propria or more advanced tumors seen at endoscopy or EUS may warrant further imaging for staging, including CT of the chest, abdomen, and pelvis and rectal MRI, depending on the location of the tumor.

Technique

Endoscopic submucosal resection first involves marking the lesion with cautery to define the boundaries of the tumor. Fluid is then injected into the submucosal space below the mass to get an adequate lift and allow dissection. The most common fluid used in the United States is hydroxypropylmethyl cellulose, which is mixed with indigo or methylene blue stain to allow quick visualization of the submucosal space. The mass is then cut around the borders with an angulated cutting edge to prevent inadvertent puncture of the muscularis or blood vessels, and the tumor is removed in one piece through the orifice. Blood vessels and tissues are then cauterized for hemostasis (14) (Fig 6). En bloc resection allows accurate histologic diagnosis and ensures that the edges of the specimen are free of tumor (Fig 7). The postresection site may then be closed with clips or endoscopic sutures, depending on the endoscopist's preference.

Table 1: Indications for Endoscopic Submucosal Dissection

Location	Indication	Surgical Alternative
Esophagus	Squamous cell carcinoma: High-grade dysplasia to well to moderately differentiated lesions Paris 0–II lesions Absolute indication: two-thirds or less involvement of the esophageal circumference Expanded indications: <200 μm involvement, any size, no regional lymphadenopathy Esophageal adenocarcinoma: Large or bulky areas of nodularity Intramucosal carcinoma Superficial submucosal invasion Recurrent dysplasia	Esophagectomy
Stomach	Absolute indications: mucosal adenocarcinoma and lesions with high-grade dysplasia), intestinal type, size 2 cm, no ulceration Expanded indications: Adenocarcinoma, intestinal type, any size, without ulceration Adenocarcinoma, intestinal type, submucosal invasion (<500 mm) Adenocarcinoma, intestinal type, size 3 cm, with ulceration Adenocarcinoma, diffuse type, size 2 cm, without ulceration	Gastrectomy
Colon or rectum	Type V Kudo pit pattern Depressed component Complex morphologic characteristics Rectosigmoid location Nongranular lateral spreading tumor (adenomas), size ≥20 mm Granular lateral spreading tumor (adenomas), size ≥30 mm Residual or recurrent colorectal adenomas	Partial colectomy Low anterior resection

Note.—Adapted, with permission from Elsevier, from references 16 and 17.

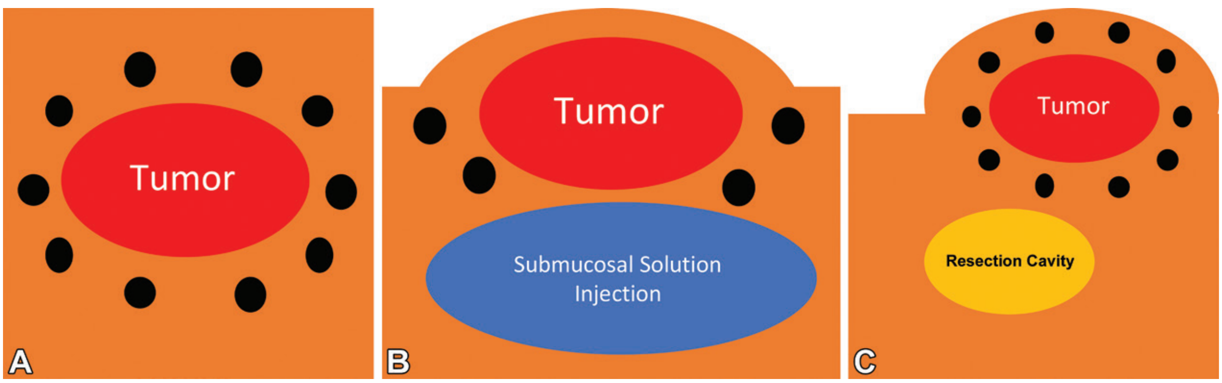


Figure 6. Illustration shows endoscopic submucosal resection of a tumor. **(A)** The boundaries of the tumor are marked with cautery. **(B)** Fluid and indigo or blue stain is injected submucosally to raise the neoplasm and allow dissection. **(C)** After adequate submucosal fluid is injected to lift the neoplasm, the tumor is cut along marked boundaries. The tumor is then removed through the orifice and all bleeding is cauterized.

Expected Postoperative Appearance

The postoperative imaging appearance after ESD is nonspecific and often normal. Common imaging findings include mild wall thickening and surrounding inflammatory changes immediately after the procedure. The inflammation shortly resolves. Clips are placed at the area of the lesion resection for hemostasis and should serve as a focal point for assessment of any potential complications at postoperative imaging (Fig 8).

Advantages and Complications

The benefits of ESD over traditional surgery are multifactorial. The main advantage of ESD over surgical intervention is that ESD is minimally invasive. Surgical methods include esophagectomy for esophageal malignancy, abdominoperineal resection with colostomy for rectal tumors, partial or total gastrectomy for gastric cancer, and partial or total colectomy for treatment of malignancies of the colon. ESD allows organ-

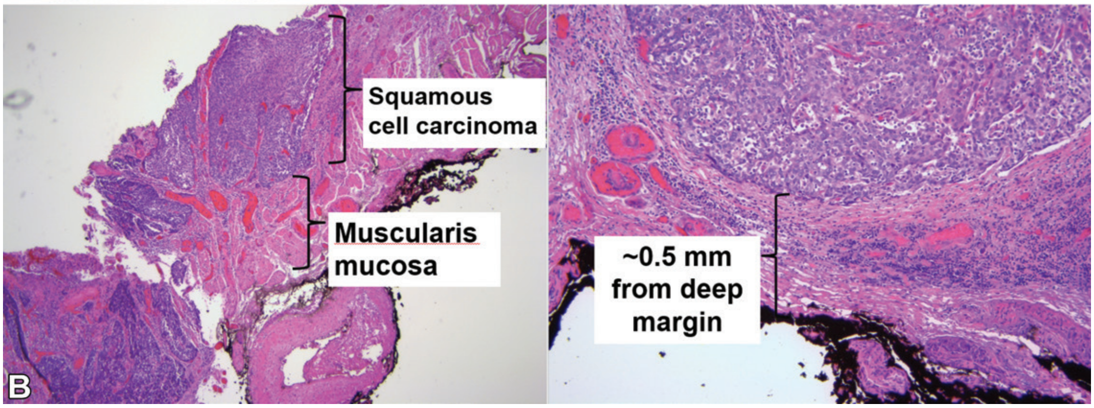


Figure 7. ESD of an esophageal neoplasm in a 56-year-old man who presented with an esophageal carcinoma that was seen on EUS images (not shown) and was confined to the mucosa without submucosal invasion, which is a finding that is consistent with a T1a neoplasm. **(A)** Photograph shows the surgical specimen from ESD with en bloc resection removed in one piece for pathologic examination. **(B)** Low-power (left) and high-power (right) photomicrographs of the surgical specimen show clear delineation of the tumor from adjacent uninvolved tissues, with 0.5 mm of clearance from the surgical margin. (Hematoxylin-eosin stain; original magnification, ×10 [left] and ×40 [right].)

Figure 8. Expected findings after endoscopic submucosal resection of a rectal tumor in a 45-year-old man who presented with rectal pain and bleeding. Axial contrast-enhanced CT image shows nonspecific rectal wall thickening (arrow). No surrounding fluid collection or free air is identified. The patient was treated conservatively and was discharged shortly thereafter to go home.

sparing intervention, with preservation of organ function in the treatment of early gastrointestinal malignancies.

When compared with surgical treatment, ESD has similar long-term curative outcomes with a high 5-year survival rate. For example, the 5-year survival rate for early esophageal malignancy reported in one study (20) was 94% for ESD versus 87.1% for esophagectomy. ESD also has shorter surgical times, shorter hospital stays, and fewer adverse events when compared with surgical intervention (rate of adverse events in patients who undergo esophagectomy, 55% vs 18.7% in those who undergo ESD) (Table 2) (20).

Complications from ESD are rare and include bleeding, perforation, fluid collections, and formation of strictures (ie, benign stricture or those that are due to recurrence of malignancy). Most patients with adverse events are treated conservatively or endoscopically, without any need for surgical intervention. CT with intravenous and oral contrast material is often performed to assess for pneumoperitoneum, leakage of oral contrast material, and fluid collections (Fig 9). Occasionally, tumors may be understaged on the basis of initial high-resolution endoscopy and EUS, warranting further evaluation with staging CT or PET/CT.

Table 2: Benefits of ESD versus Surgical Intervention
Comparable 5-year survival rates
Organ-sparing intervention, with preservation of function
Fewer adverse events
Shorter surgery time
Shorter hospital stay
Source.—Reference 20.

Figure 9. Acute abdominal pain in a 74-year-old woman who presented shortly after undergoing ESD of an ascending colonic mass during which a pneumoperitoneum was found. Axial CT image with intravenous and oral contrast material shows multiple locules of air (arrows) surrounding the cecum and ascending colon (circle), which is consistent with a perforation. No extraluminal oral contrast material was noted. The patient was treated conservatively.

Endoscopic Bariatric Therapy

Indications, Advantages, and Preliminary Workup

Obesity is a global health problem, affecting more than 600 million people worldwide (21). Although different strategies have been promoted to address the problem, the evidence shows that bariatric surgery provides the most effective treatment, sustained weight loss, and mitigation of comorbidities (22–26). Surgical options include laparoscopic and open Roux-en-Y gastric bypass, ESG, placement of an adjustable gastric band, vertical banded gastroplasty, duodenal switch surgery, and biliopancreatic diversion. The steady increase in the number of bariatric procedures performed opens the door for innovation, especially given that fewer than 2% of eligible patients undergo bariatric surgery because of surgical and morbidity risks, cost, access, and patient preference (27,28) Although the mortality rates for bariatric surgery are low (ie, a reported rate of 0.08% within 30 days after surgery and 0.31% after 30 days, according to a meta-analysis published in 2014 [29])and the complication rate is decreasing (10.5% in 1993, 7.6% in 2006 [30]), endoscopic techniques offer a safer and more cost-effective alternative (31,32).

Endoscopic bariatric therapy is an effective option for treatment of obesity in patients with a body mass index of 30–40 kg/m^2. Endoscopic bariatric therapy can also be used as a bridge to allow patients to undergo surgical procedures for which they do not meet weight requirements, such as orthopedic or transplant surgery (33–35). Several endoscopic procedures have been developed. The two most common techniques in the United States are placement of an intragastric balloon and ESG (7,36). Before placement of an intragastric balloon, endoscopy is typically performed to exclude the presence of a large hiatal hernia. No consensus exists on preprocedural imaging for placement of a gastric balloon (37). In patients who have undergone prior surgical procedures, preprocedural fluoroscopic imaging may be performed to evaluate for anatomic changes of the gastrointestinal tract.

Endoscopic Technique

The goal of placement of an intragastric balloon is to facilitate weight loss by inducing early satiety through gastric distention, reducing production of hunger-inducing hormones, and slowing gastric emptying. These devices are typically placed endoscopically for a duration of 4–12 months and are inflated with 450–700 mL of saline solution and methylene blue stain, which appears in the urine when a balloon ruptures. Some balloons (Obalon Balloon System; Obalon Therapeutics) are also inflated with a gas. Use of gas-filled intragastric balloons requires placement of 2–3 balloons to achieve early satiety, given their smaller size. Most intragastric balloons are removed endoscopically, but they also have a spontaneous deflation rate of 3%–23%. The Elipse balloon (Moore Metabolics) is an exception, because it is excreted (38). Three intragastric balloons are currently approved by the U.S. Food and Drug Administration, with several pending or still in the approval process (39). Contraindications to intragastric balloons include a large hiatal hernia, prior gastric or bariatric surgery, inflammatory

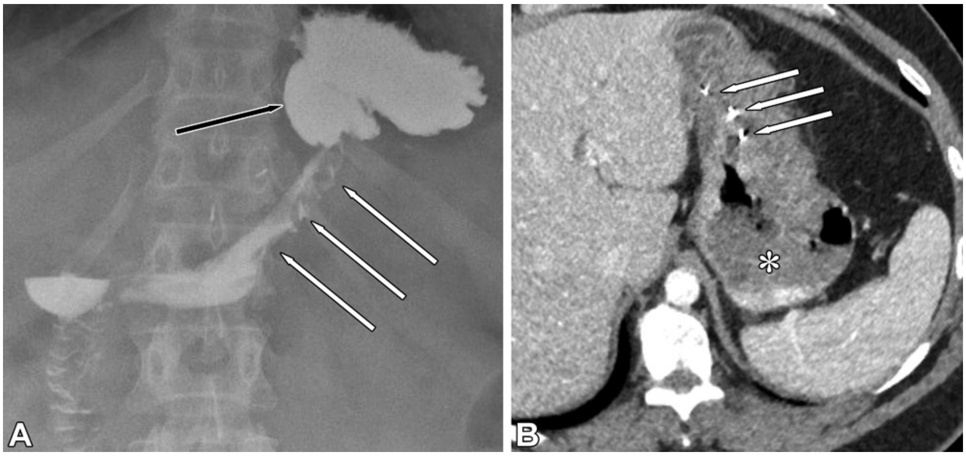


Figure 10. Imaging appearance after ESG in two women. **(A)** Anteroposterior fluoroscopic image of the stomach in a 32-year-old woman after administration of oral contrast material shows metallic structures (white arrows) that anchor and cinch the greater curvature of the stomach, creating gastric luminal shortening and narrowing to promote early satiety. Note the spared fundus (black arrow), which is an expected postoperative finding and should not be mistaken for failure of ESG. **(B)** Axial contrast-enhanced CT image through the upper abdomen in a 41-year-old woman shows the expected appearance after ESG, with sutures along the greater curvature of the stomach (arrows) and narrowing of the gastric lumen to promote early satiety via restriction. Note the sparing of the gastric fundus (*), which remains open as a reservoir.

bowel disease, and an increased risk of upper gastrointestinal bleeding (40).

ESG is performed with an endoscopic suturing system to constrict the gastric lumen. The goal of ESG is to reduce gastric intake through restriction and induce subsequent weight loss. Transmural radiopaque suture anchors are placed in a triangular configuration longitudinally along the anterior and posterior walls of the greater curvature of the stomach, sparing the fundus, to constrict the gastric lumen, reducing gastric width and length to create a smaller stomach volume. The fundus is not sutured because doing so results in an increased risk of leakage. As opposed to its laparoscopic counterpart, ESG is incisionless and maintains the integrity of the gastric structure, facilitating revision or conversion to an open bariatric procedure, if necessary.

Expected Postoperative Appearance

Abdominal radiographs are the standard imaging method to identify and locate the intragastric balloon and radiopaque valve. The balloon is readily apparent on CT images and appears as a round or elliptical structure with attenuation similar to that of gas or water, depending on whether it is inflated with gas or saline solution. The valve is normally positioned in the gastric fundus with a metallic nidus conforming to the valve. ESG can be followed by fluoroscopic imaging of the upper gastrointestinal system when there is clinical concern for a leak. The gastric lumen appears constricted and tubular, sparing the fundus, which is flared outward and should not be misinterpreted as dehiscence or a suboptimal postprocedural appearance. Preserving the normal fundus also helps to prevent gastroesophageal reflux and enhances satiety by serving as a food reservoir (41,42). Gastric transit is typically delayed because of postprocedural edema, which is reflected by the thickened rugal folds. Familiarity with the typical plicated appearance of the gastroplasty site is necessary to differentiate the expected appearance of gastroplasty from that of a small leak. Small protrusions of contrast material through the gastric plications is expected, especially in the left lateral decubitus position, but no opacification of the plicated stomach should be visualized. Because of the possibility of peritoneal leakage, postprocedural fluoroscopy should be performed with water-soluble contrast material, after which barium can be administered if there is no leak (Fig 10). Any post-ESG fluoroscopic evaluation, particularly when the procedure was performed in a more remote location, should include assessment of the integrity of the sutures and the degree of gastric restriction.

Post-ESG radiographs show the metal anchors and cinching of the gastroplasty and occasionally show a small amount of pneumoperitoneum. The actual sutures used in ESG are made of prolene and are not seen radiographically. CT shows edematous wall thickening, the metallic anchors and cinching along the gastroplasty site, and possibly, a small amount of pneumoperitoneum, with absence of the fluid and fat stranding seen with surgical procedures (Fig 10).

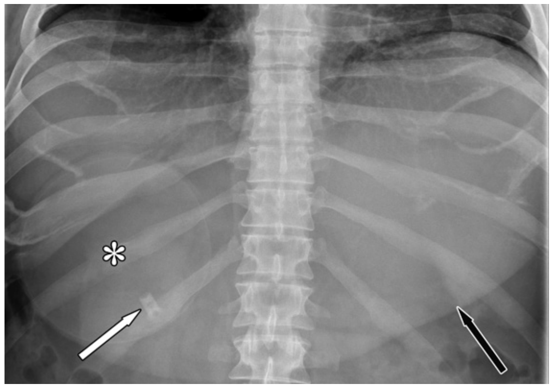


Figure 11. Gastric outlet obstruction from a migrated intragastric balloon in a 33-year-old woman with nausea, vomiting, and abdominal pain. Frontal radiograph shows massive gastric distention (black arrow) due to migration of the intragastric balloon (*) to the gastric antrum. Note the position of the valve projecting into the lumen of the balloon (white arrow).

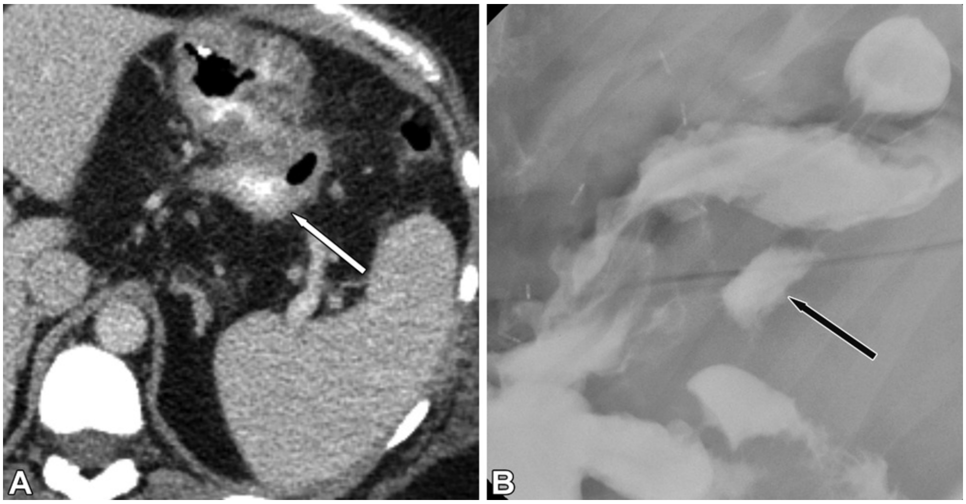


Figure 12. Gastric leak in a 29-year-old man with abdominal pain after ESG. **(A)** Axial CT image through the gastric body with intravenous and oral contrast material shows an extraluminal collection of oral contrast material (arrow) posterior to the greater curvature of the stomach, which is suspicious for perforation and leak. **(B)** Anteroposterior fluoroscopic image of the stomach shows an extraluminal collection of oral contrast material adjacent to the greater curvature of the stomach (arrow), which is consistent with a postoperative leak.

Complications

Common adverse events of intragastric balloons include abdominal pain, nausea, vomiting, and dyspepsia (40). An early and rare complication of the use of intragastric balloons is gastric perforation. However, modifications to these prior models have minimized the ulcerogenic potential of these devices, reducing the incidence of ulcers from 35% to 10% (43). Any amount of pneumoperitoneum is abnormal and not expected after intragastric balloon deployment. Balloons can deflate over time, leading to bowel obstruction with distal migration, and deflation can be assessed with US (44). Migration of the balloon poses the risk of gastric outlet obstruction, which is seen radiographically as abnormal positioning of the (usually inflated) balloon near the pylorus in the right upper quadrant, with a distended gastric shadow (Fig 11).

ESG has a lower complication rate than that of surgery, with Fayad et al (45) reporting an adverse event rate of 4.76%, compared with 12.31% for laparoscopic gastroplasty, in a 2018 retrospective case-control study. In a recent review (46) of nine ESG studies, a rate of adverse events of 2.3% was found and included leaks, perigastric fluid collections, pulmonary embolism, and pneumoperitoneum, with pneumothorax. Patients with adverse events such as abdominal pain, nausea, and emesis are treated conservatively and show improvement after few days. A small amount of pneumoperitoneum is not unexpected and usually does not indicate gastric perforation, which is rare but has been reported (47,48). More than a small quantity of pneumoperitoneum may suggest the possibility of gastric perforation but should be evaluated with fluoroscopy or CT with oral contrast material to assess for leakage of contrast material. The challenges to diagnosis of perforation are to discriminate between extraluminal contrast material and contrast material that is outlining

Figure 13. Suture dehiscence and perforation after ESG in a 37-year-old man who presented with a fever. Contrast-enhanced axial CT image of the upper abdomen shows a large perigastric collection of fluid (*) adjacent to the gastric fundus. Multiple locules of perihepatic pneumoperitoneum are present (arrow), which are consistent with suture dehiscence and perforation.

plications between gastroplasty sutures (Fig 12). Dependent positioning on the left side during fluoroscopy facilitates passage of contrast material outside the confines of the stomach in the case of perforation or into the plicated stomach in the case of suture dehiscence. Perigastric fluid collections are also rare complications seen on CT or fluoroscopic images and are often associated with pneumoperitoneum (Fig 13).

EUS-guided Drainage through LAMS and Covered Metal Stents

Overview and Preoperative Imaging

The use of covered metal stents and LAMS has expanded from the drainage of fluid collections in patients with pancreatitis to biliary or gallbladder drainage, drainage of nonpancreatically related fluid collections, and creation of gastrojejunostomy in patients with gastric outlet obstruction (3,49). These techniques are advantageous in patients who would be difficult to access for interventional radiology or whose condition is too unstable for surgery. In addition, EUS-guided therapy has allowed endoscopists to treat patients while avoiding discomfort and potential dislodgement of percutaneous drains and allowing lower costs and complication rates when compared with those of surgical intervention. EUS gives endoscopists the ability to find the target in real time while avoiding surrounding structures such as vessels and intervening organs.

EUS-directed drainage procedures gain access to the target lesion through fully covered metal stents or LAMS, which are both larger in caliber than plastic stents, allowing a longer length of patency with less stent occlusion. LAMS are dumbbell-shaped fully covered metal stents that are available in diameters of 10 mm, 15 mm, and 20 mm. The dumbbell shape decreases the risk of migration, and the larger diameter allows the passage of an endoscope through the LAMS to perform endoscopic therapy (50). The deployment system is cautery-enhanced and allows direct puncture and deployment of the stent without the need to exchange instruments. Stents are placed through a transluminal approach with creation of a fistula between the hollow viscous organ and target lesion under EUS guidance. Once the LAMS is deployed, contrast material can be injected to confirm the location of the stent.

Preoperative imaging depends on the clinical scenario or indication and includes contrast-enhanced CT to assess for bowel and/or altered gastric anatomy, gastric outlet obstruction and collections, and MRI and MR cholangiopancreatography to determine the level of biliary obstruction. The limitations of EUS-directed procedures include the lack of widespread multi-institutional expertise, the increased invasiveness when compared with percutaneous intervention, and the limitation of its use to structures that are in close proximity to the stomach, esophagus, and bowel (51).

Endoscopic Techniques and Indications

EUS-guided Biliary Drainage.—Indications for EUS-directed biliary drainage include patients who have gastric outlet obstruction or surgically altered anatomy, such that EUS-directed transgastric endoscopic retrograde cholangiopancreatography (EDGE) cannot be performed (52). Options for EUS-directed biliary drainage in patients with gastric outlet and biliary obstruction include EUS-guided placement of antegrade transpapillary stents and EUS-guided retrograde placement of a transpapillary stent (ie, the "rendezvous" procedure) (53). Biliary drainage can also be achieved through EUS-guided choledochoduodenostomy, where the common bile duct is drained from the duodenal bulb, and EUS-guided hepaticogastrostomy, where the left intrahepatic ducts are drained from the stomach (54). EUS-guided choledochoduodenostomy and hepaticogastrostomy are performed by advancing an echoendoscope into the duodenal bulb or stomach, respectively, and identifying the dilated bile ducts. Doppler flow US is used to differentiate vessels from adjacent bile ducts. The duct is then punctured with a

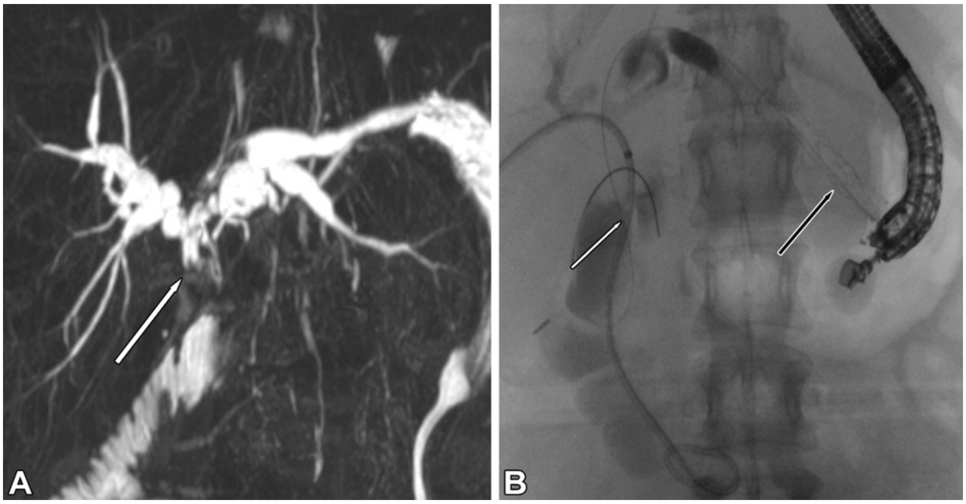


Figure 14. Anastomotic stricture in a 43-year-old man who underwent Roux-en-Y hepaticojejunostomy for a type 1 choledochal cyst. EUS-guided hepaticogastrostomy through the left hepatic ducts was performed to help drain the biliary system. **(A)** Three-dimensional maximum intensity projection MR cholangiopancreatographic image shows marked intrahepatic biliary ductal dilatation with central tapering (arrow), which is consistent with a stricture at the hepaticojejunostomy site. Note the proximity of the dilated left intrahepatic bile ducts to the adjacent stomach. **(B)** Fluoroscopic image obtained during EUS-guided hepaticogastrostomy shows placement of a fully covered metal stent between the stomach and the dilated left intrahepatic bile ducts (black arrow). A guidewire (white arrow) was advanced through the stent across the hepaticojejunal anastomosis to maintain access, dilate the anastomotic stricture with a balloon, and guide the placement of a stent.

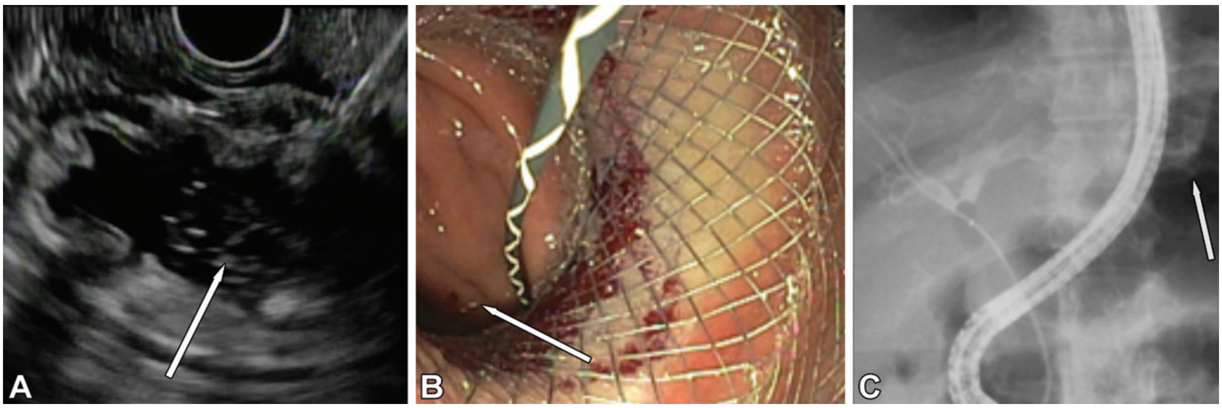


Figure 15. Biliary obstruction from a common bile duct stone in a 22-year-old woman who had undergone Roux-en-Y bypass for weight loss. EUS-directed transgastric endoscopic retrograde cholangiopancreatography (EDGE) was performed to remove a common bile duct stone. **(A)** EUS image shows the excluded stomach through the gastric pouch (arrow), which is accessed with a fine needle for deployment of the LAMS. **(B)** Endoscopic view after placement of the LAMS shows direct connection of the excluded stomach (arrow). **(C)** Endoscopic access was achieved through the placed LAMS (arrow) into the excluded stomach, with opacification of the biliary system.

fine needle, and a cholangiogram is acquired, along with fine-needle aspiration, to confirm the location. A guidewire is advanced through the fine needle into the bile duct, and a cautery-enhanced needle knife or cystotome is advanced over the wire to enlarge the opening and deploy a fully covered metal biliary stent into the bile duct, with the proximal end in either the duodenal bulb or stomach (Fig 14) (55). The placement of the stent in the bile ducts is confirmed with fluoroscopic injection of contrast material.

Biliary access in patients with obstructed bile ducts and Roux-en-Y anatomy can be achieved with an EDGE procedure. EDGE involves the use of EUS to visualize the bypassed stomach. The bypassed stomach is accessed via the EUS-directed fine needle, through which saline solution is instilled to distend the excluded stomach (56). A LAMS is deployed to create a gastrogastric fistula with the use of cautery. The LAMS is dilated to at least 15 mm, and the duodenoscope can then be inserted through the LAMS into the bypassed stomach to perform an EDGE procedure (Fig 15). The stent is left in place for at least 2–3 weeks until access is no longer needed, after which the LAMS is removed, and

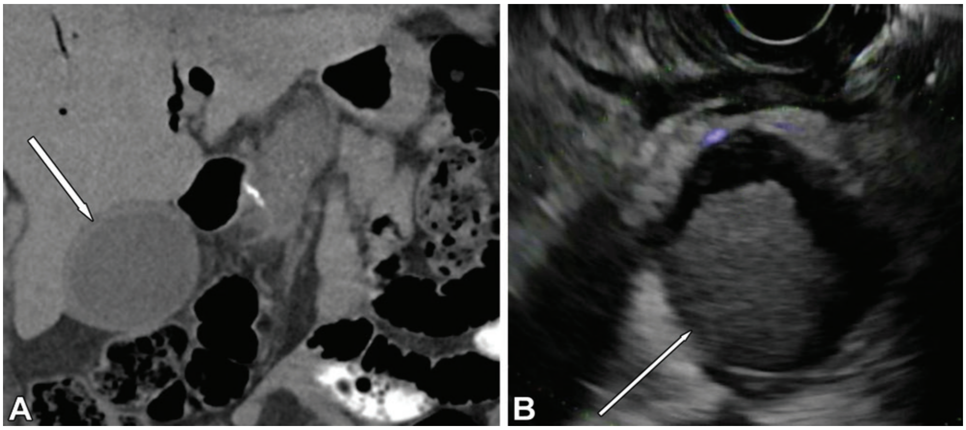


Figure 16. Acute cholecystitis and multiple comorbidities that preclude surgery in an 85-year-old woman who underwent EUS-guided drainage into the duodenum. **(A)** Coronal noncontrast CT image of the abdomen shows a distended gallbladder with wall thickening (arrow). **(B)** EUS image shows the inflamed gallbladder (arrow), which is filled with sludge. The gallbladder was accessed and a LAMS was deployed.

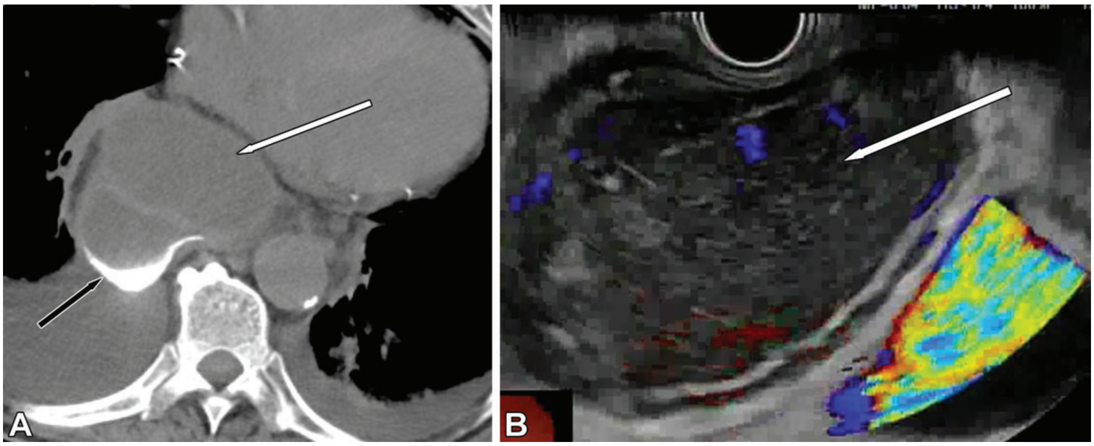


Figure 17. Mediastinal fluid collection in a 55-year-old man who underwent recent esophagectomy for malignancy and presented with a fever. **(A)** Axial noncontrast CT image of the chest shows a collection of oral contrast material (white arrow) in the mid mediastinum between the reconstructed area of the esophagus (ie, "gastric pull-up") (black arrow) and the heart. Small pleural effusions are greater on the right side than those on the left. The collection was not amenable to percutaneous drainage, and the patient's condition was too unstable for surgical intervention. **(B)** EUS image shows the mediastinal fluid collection (arrow) with heterogeneous hypoechoic debris and no internal flow. The collection was accessed by using a 19-gauge needle (not shown), and a LAMS was deployed. Purulent fluid was elicited from the stent at endoscopy, with drainage into the gastric pull-up.

the fistula tract is closed by clipping or suturing endoscopically (57).

EUS-directed Drainage of the Gallbladder and Abscesses.—EUS-directed drainage of the gallbladder and abscesses are alternatives for patients with acute cholecystitis or fluid collections who have contraindications for surgery or are not good candidates for percutaneous drainage because of the location of the fluid collection or the gallbladder (58,59). Drainage of the gallbladder and nonpancreatically related fluid collections through an endoscopic LAMS is performed similarly by first visualizing the fluid-filled structure with EUS through the adjacent bowel (60,61). Doppler US is used to exclude intervening vessels. A 19-gauge fine needle is inserted into the gallbladder or collection; a guidewire may be used to maintain access. The LAMS is then placed by first deploying the internal flange in the fluid collection or gallbladder, and the external flange is deployed in the lumen of the gastrointestinal tract (Figs 16, 17). Any fluid collection that is in close proximity to the bowel can be accessed through this method and drained via a LAMS (62).

EUS-directed Gastrojejunostomy.—A gastrojejunal bypass can be performed endoscopically in patients with gastric outlet obstruction due to malignancy as an alternative to a duodenal stent or surgical gastrojejunostomy. EUS-directed gastrojejunostomy is performed by first inserting a transnasal endoscopically guided wire and

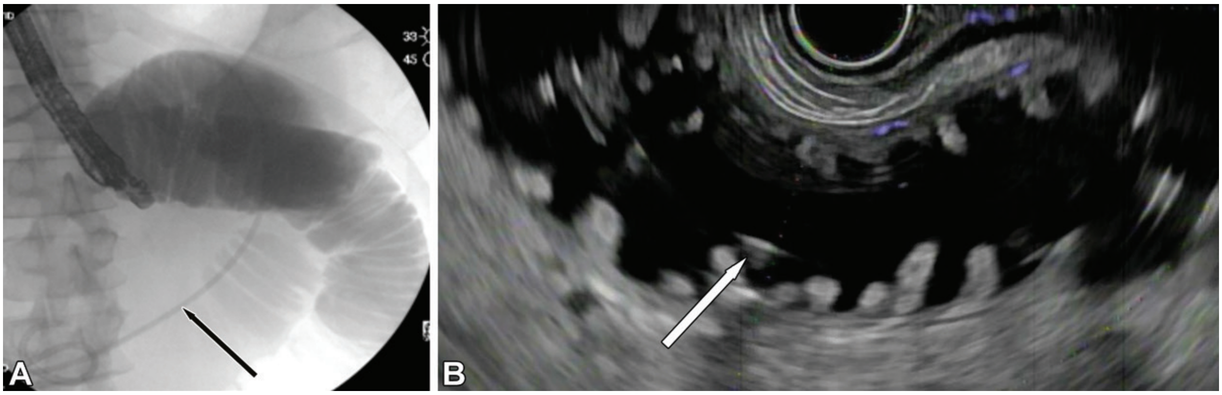
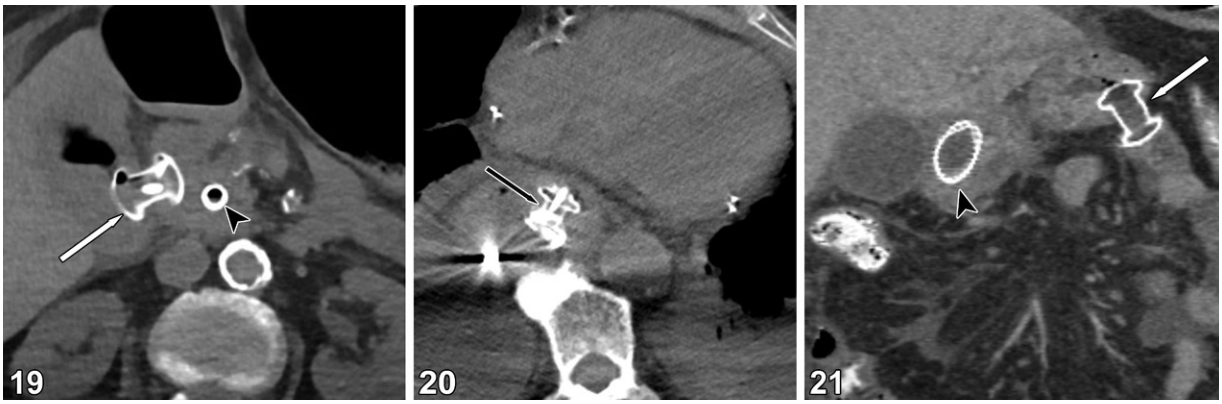


Figure 18. Endoscopic gastrojejunostomy performed in a 46-year-old man with gastric outlet obstruction from a pancreatic malignancy. **(A)** Fluoroscopic image shows that the jejunum is accessed via a transnasal endoscopically guided catheter, and the jejunum is extended with normal saline solution mixed with methylene blue stain. **(B)** Transverse US image using an echoendoscope shows the distended jejunum containing the catheter for deployment of the LAMS through the stomach into the adjacent small bowel. The distended loop of jejunum is accessed from the stomach with a 19-gauge needle. With the use of the guidewire, the LAMS is deployed.



Figures 19–21. **(19)** Expected appearance of EUS-directed gallbladder drainage in a 79-year-old woman with a pancreatic malignancy. Axial oral contrast material–enhanced CT of the abdomen shows a decompressed gallbladder, with air after placement of a LAMS (arrow). Access to the gallbladder was obtained through the duodenum. Note the common stent in place in the bile duct (arrowhead). **(20)** Normal postoperative appearance in a 39-year-old man after EUS-directed placement of a LAMS from a gastric pull-up into a mediastinal fluid collection. Axial noncontrast CT image shows a LAMS (arrow) extending from the gastric pull-up into a mediastinal fluid collection. Note the moderate bilateral pleural effusions with bibasilar atelectasis. **(21)** EUS gastrojejunal bypass in a 47-year-old man with gastric outlet obstruction from a pancreatic malignancy. Axial contrast-enhanced CT image shows the expected posttreatment appearance of a gastrojejunal bypass, with a LAMS connecting the greater curvature of the stomach to a nearby jejunal loop (arrow). The patient also has a duodenal stent (arrowhead) that was placed because of gastric outlet obstruction from a pancreatic adenocarcinoma.

catheter into the jejunum to distend the small bowel with normal saline solution mixed with methylene blue stain. An echoendoscope is then inserted into the stomach to visualize the distended loop of jejunum containing the catheter (63). The loop of jejunum is accessed through a fine needle, with confirmation obtained if blue saline solution is aspirated. The LAMS delivery system is then advanced into the jejunum using cautery, and the LAMS is deployed (Fig 18).

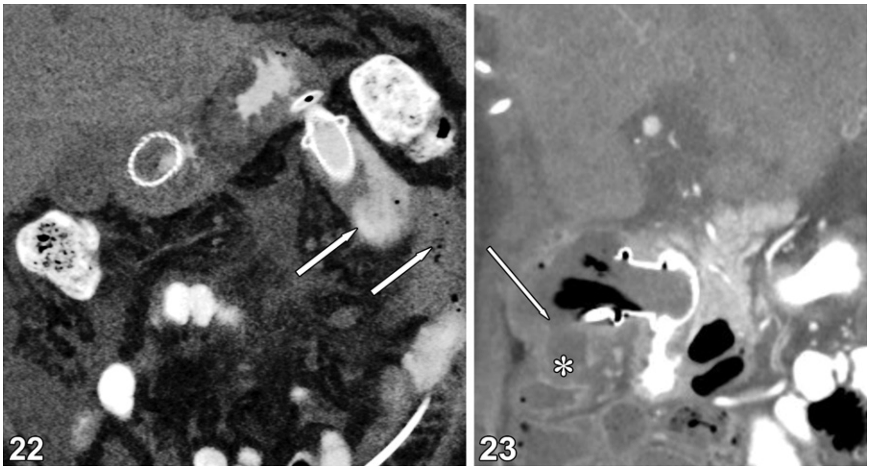
Expected Postoperative Appearance

The postprocedural appearance at CT and MRI in patients who have undergone EUS-guided drainage procedures demonstrates a LAMS or covered metal stent connecting the bowel and the target lesion, depending on what procedure was performed. EUS-guided drainage of the gallbladder shows a LAMS connecting the antrum or the duodenum and the gallbladder (Fig 19). Drainage of an abscess at CT or MRI shows a LAMS connecting the closest adjacent bowel segment to the fluid collection (Fig 20). Endoscopically guided gastrojejunostomy at CT shows a LAMS connecting the greater curvature of the stomach and an adjacent loop of the jejunum (Fig 21). Drainage of the common bile duct and the left hepatic ducts shows a covered metal stent between the duodenum to the common bile duct and stomach to the left intrahepatic ducts, respectively (Fig 14). Pneumoperitoneum is often a common and expected finding immediately after the procedure. On all postprocedural images, the stent must be care-

Table 3: EUS-directed Drainage Procedures—Efficacy and Complication Rates

EUS-directed Procedure	Technical Success Rate (%)	Clinical Success Rate (%)	Complication Rate (%)
Gallbladder drainage	81–96	75–88	3.6–6.3
Gastrojejunostomy	92	85	11.5
EDGE	100	91	6.25
Abscess drainage	93.3–100	75–100	0–25
EUS-directed biliary drainage	64.7–96.2	63.2–100	10–35

Sources.—References 51, 57, and 63.



Figures 22, 23. **(22)** Leak in a 42-year-old woman with an EUS gastrojejunal bypass and malignant gastric outlet obstruction. Coronal oral contrast–enhanced CT image shows leakage of oral contrast material from a LAMS gastrojejunostomy bypass, with adjacent locules of free air (arrows), which are consistent with a leak or perforation. **(23)** Leak and abscess in a 59-year-old man with EUS-directed gallbladder drainage. Coronal contrast-enhanced CT image shows an EUS-directed LAMS in place for drainage of the gallbladder. A defect in the posterior wall of the gallbladder (arrow) and a fluid collection (*) have developed, which are findings consistent with gallbladder perforation and abscess.

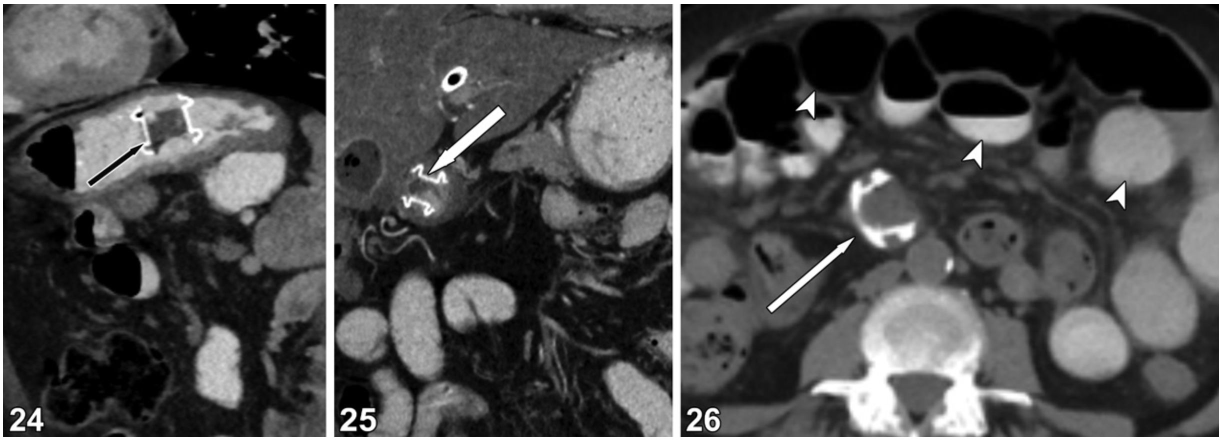
fully assessed for subtle changes in placement that could suggest migration.

Advantages and Complications

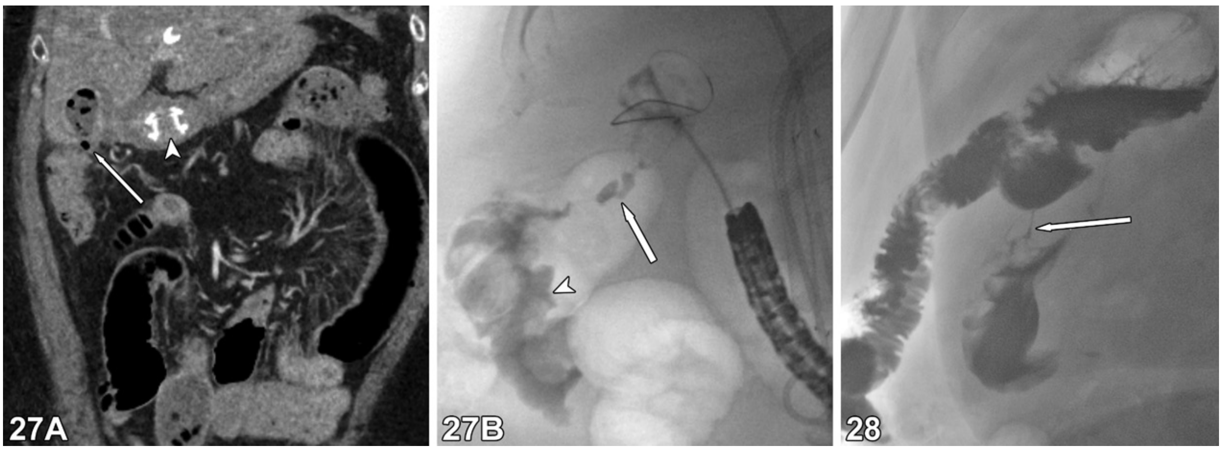
EUS-guided drainage procedures have excellent technical and clinical success rates, with an overall low rate of adverse events (Table 3). Because of the minimally invasive nature of these procedures, they often can be performed in patients for whom surgery is contraindicated and those in whom a percutaneous approach is not amenable (64). Serious adverse events are rare, occurring in 10%–20% of cases, and the majority of these adverse events are mild (Table 3). Complications include fluid collections or abscesses, bile leaks, hemoperitoneum, stent migration, stent occlusion, and perforation (65). Abscesses or fluid collections are a result of leakage around the stent and manifest at imaging as greater than expected pneumoperitoneum with spillage of extraluminal contrast material and fluid surrounding the stent (Figs 22, 23). Stent migration is one of the most common adverse events and can result in recurrent biliary obstruction, bile leaks, or bowel obstruction, depending on which procedure was performed (Figs 24–26) (66). Leakage around stents between the bowel and the gallbladder or bile ducts can result in biliary leaks, which can result in fistulas with adjacent structures owing to the caustic nature of bile (Fig 27). A specific complication that is associated with the EDGE procedure after removal of the LAMS is a persistent patency of the fistula tract between the excluded stomach and gastric pouch, resulting in a gastrogastric fistula and recurrent weight gain (Fig 28).

Conclusion

Advanced endoscopic techniques are important in the treatment of patients with gastrointestinal and biliary abnormalities and are becoming more common in the hospital setting. The benefits of these procedures compared with surgery are substantial, because they are minimally invasive, allow organ preservation, can be performed in patients for



Figures 24–26. **(24)** Migration of a LAMS in a 28-year-old man who underwent gastric bypass with need for biliary access. Sagittal contrast-enhanced CT image shows a LAMS from a prior EDGE procedure that has migrated into the excluded stomach (arrow). **(25)** Migration of a LAMS that was placed for EUS-directed gallbladder drainage in a 41-year-old man. Coronal contrast-enhanced CT image acquired several months after placement of a LAMS for cholecystitis shows migration of the stent, which is now in the duodenum (arrow). **(26)** Migration of a LAMS that was placed for an EDGE procedure, with resultant small bowel obstruction, in a 40-year-old woman. Axial oral contrast material–enhanced CT image of the abdomen shows dilated loops of small bowel (arrowheads) secondary to migration of a LAMS (arrow) to the distal ileum, with a distally decompressed bowel.



Figures 27, 28. **(27)** Migration of a LAMS placed for gallbladder drainage in an 87-year-old man, with development of cholecystocolonic fistula to the hepatic flexure. **(A)** Coronal oral contrast-enhanced CT image shows migration of a LAMS into the duodenum (arrowhead). Foci of extraluminal gas between the hepatic flexure and gallbladder can be seen, with associated colonic wall thickening that is suspicious for a potential fistula. **(B)** Single fluoroscopic image obtained at endoscopic retrograde cholangiopancreatography, which was performed for assessment of a gallbladder leak, shows a fistula (arrow), extending from the gallbladder to the hepatic flexure, with colonic opacification (arrowhead). **(28)** Gastrogastric fistula in a 33-year-old woman with recurrent weight gain and a history of gastric bypass and an EDGE procedure who underwent removal of a LAMS. Fluoroscopic image shows a fistulous connection (arrow) between the gastric pouch or proximal efferent limb and the excluded stomach, which is consistent with a gastrogastric fistula.

whom more invasive procedures are contraindicated, result in a shorter hospital stay, and involve fewer postoperative complications. Radiologists play an important role in the imaging of patients with gastrointestinal and biliary abnormalities and must be familiar with the relevant terminology and recognize the normal imaging appearance after these procedures. In addition, radiologists must be able to identify complications including leaks, abscesses, stent migration, and perforation to communicate with ordering clinicians effectively.

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